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Build a Complete System Hardware Profile

System vendor	: VMware, Inc.
Product/model	: VMware Virtual Platform
Motherboard/baseboard	: Intel corporation, 440BX Desktop Reference Platform
BIOS version/date	: Version 6.00, 11/20/2020
Kernel version	: Linux 6.17.0-19-generic
Architecture type	: x86_64
Machine	: Virtual

Informasi yang berasal dari firmware table adalah system vendor, product/model, motherboard/baseboard, BIOS version/date. Sedangkan informasi tentang kernel didapat dari uname-a. Dapat dilihat dari system vendor dan juga product mengindikasikan bahwa system linux berasal dari virtual machine.

System vendor (manufacturer) & Product/model (product name)

```
fakhruy@ubuntudesktop01:~$ sudo dmidecode -t system
# dmidecode 3.5
Getting SMBIOS data from sysfs.
SMBIOS 2.7 present.

Handle 0x0001, DMI type 1, 27 bytes
System Information
    Manufacturer: VMware, Inc.
    Product Name: VMware Virtual Platform
    Version: None
    Serial Number: VMware-56 4d 2f 74 10 6c e1 9f-80 07 4d 60 a4 b3 4e 7b
    UUID: 742f4d56-6c10-9fe1-8007-4d60a4b34e7b
```

Motherboard/baseboard

```
fakhruy@ubuntudesktop01:~$ sudo dmidecode -t baseboard
# dmidecode 3.5
Getting SMBIOS data from sysfs.
SMBIOS 2.7 present.

Handle 0x0002, DMI type 2, 15 bytes
Base Board Information
    Manufacturer: Intel Corporation
    Product Name: 440BX Desktop Reference Platform
    Version: None
    Serial Number: None
    Asset Tag: Not Specified
```

BIOS version/date

```
fakhrusy@ubuntudesktop01:~$ sudo dmidecode -t bios
# dmidecode 3.5
Getting SMBIOS data from sysfs.
SMBIOS 2.7 present.

Handle 0x0000, DMI type 0, 24 bytes
BIOS Information
    Vendor: Phoenix Technologies LTD
    Version: 6.00
    Release Date: 11/12/2020
    Address: 0xEA480
    Runtime Size: 88960 bytes
    ROM Size: 64 kB
    Characteristics:
```

Kernel version & Architecture type

```
fakhrusy@ubuntudesktop01:~$ uname -a
Linux ubuntudesktop01 6.17.0-19-generic #19~24.04.2-Ubuntu SMP PREEMPT_DYNAMIC Fri Mar  6 23:08:46 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
fakhrusy@ubuntudesktop01:~$
```

Investigate CPU Architecture and Processing Topology

Architecture : x86_64
CPU op-mode : 32-bit, 64-bit
Sockets : 2
Cores per socket : 2
Threads per core : 1
Model name : 12th Gen Intel(R) Core(TM) i7-12700H
Vendor ID : Genuine Intel
Flags : pu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat
pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon
rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3
fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor
lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2
erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni

arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrc md_clear serialize flush_lld
arch_capabilities

Logical CPUs : sockets (2) x cores per socket (2) x threads per core (1) = 4 (true, verify dengan nproc)

Perbedaan processor, core, thread, dan socket. Socket adalah slot atau wadah fisik tempat CPU dipasang. Processor atau CPU merupakan chip yang dipasang di dalam socket. Core adalah komponen di dalam CPU yang bertugas untuk memproses data dan instruksi. Thread adalah unit eksekusi terkecil dari proses di dalam core CPU.

Linux melihat CPU logis sebagai suatu unit yang dapat dijadwalkan. Tidak ada perbedaan dalam penjadwalan, sehingga semua dianggap sebagai resource yang dapat menjalankan tugas.

```
fakhrusy@ubuntudesktop01:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          45 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 4
On-line CPU(s) list:   0-3
Vendor ID:              GenuineIntel
Model name:             12th Gen Intel(R) Core(TM) i7-12700H
CPU family:             6
Model:                 154
Thread(s) per core:     1
Core(s) per socket:     2
Socket(s):              2
Stepping:               3
BogoMIPS:               5375.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss h
t syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_
tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave
avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves av
x_vnni arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrc md_clear serialize flush_lld arch_c
apabilities
```

```
fakhrusy@ubuntudesktop01:~$ nproc
4
fakhrusy@ubuntudesktop01:~$ nproc
4
fakhrusy@ubuntudesktop01:~$ nproc
4
fakhrusy@ubuntudesktop01:~$ nproc
4
```

```

processor      : 0
vendor_id     : GenuineIntel
cpu_family    : 6
model         : 154
model name    : 12th Gen Intel(R) Core(TM) i7-12700H
stepping      : 3
microcode     : 0xffffffff
cpu MHz       : 2687.999
cache size    : 24576 KB
physical id   : 0
siblings      : 2
core id       : 0
cpu cores     : 2
apicid        : 0
initial apicid : 0
fpu           : yes
fpu_exception : yes
cpuid level   : 32
wp            : yes
flags         : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall n
x pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs i
bpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsave
s avx_vnni arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_lid arch_capabilities
bugs          : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs retbleed rfds bhi spectre_v2_user its
bogomips      : 5375.99
clflush size   : 64
cache_alignm   : 64
address sizes  : 45 bits physical, 48 bits virtual
power management:

```

Examine CPU Flags, Virtualization Support, and Instruction Capabilities

Tidak ditemukan vmx maupun svm setelah saya memasukkan command yang diminta. Salah satu kemungkinan penyebab hal tersebut dapat terjadi adalah tidak diaktifkannya nested virtualization, yang menyebabkan VM tidak dapat melihat kemampuan hardware virtualization.

AES (Advanced Encryption Standard) berfungsi untuk mengenkripsi dan mendekripsi data sensitif. Keuntungan dari fitur ini adalah salah satunya penggunaan VPN.

AVX (Advanced Vector Extensions) berfungsi untuk mempercepat komputasi berat, seperti pemrosesan audio, video, AI, dan juga aplikasi yang membutuhkan perhitungan vektor/matematika kompleks.

SSE4 (Streaming SIMD Extensions 4) berfungsi untuk mempercepat pemrosesan data tertentu.

```

fakhrusy@ubuntudesktop01:~$ egrep -o 'vmx|svm|aes|avx|avx2|sse4_1|sse4_2' /proc/cpuinfo | sort | uniq
aes
avx
avx2
sse4_1
sse4_2
fakhrusy@ubuntudesktop01:~$ grep -m1 '^flags' /proc/cpuinfo
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall n
x pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs i
bpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsave
s avx_vnni arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_lid arch_capabilities
fakhrusy@ubuntudesktop01:~$

```


Analyze CPU Cache Hierarchy and Its Relationship to Performance

L1, data, 48k, 0 (private)

L1, instruction, 32k, 0 (private)

L2, unified, 1280k, 0 (private)

L3, unified, 24576k, 0-1 (shared)

Cache L1 terintegrasi langsung di dalam CPU, sehingga jarak fisiknya lebih pendek yang memungkinkan akses data lebih cepat. Ukuran yang lebih kecil memungkinkan pencarian data (latensi) yang lebih cepat. Cache L3 letaknya lebih jauh dari inti core, sehingga latensi data lebih besar. L3 juga digunakan untuk berbagi data antar core untuk mengurangi akses kepada RAM yang lebih lambat.

```
fakhrusy@ubuntudesktop01:~$ lscpu -C
NAME ONE-SIZE ALL-SIZE WAYS TYPE          LEVEL  SETS  PHY-LINE COHERENCY-SIZE
L1d    48K    192K    12 Data              1     64      1         64
L1i    32K    128K     8 Instruction       1     64      1         64
L2     1.3M     5M    10 Unified           2    2048      1         64
L3     24M    48M    12 Unified           3   32768      1         64
fakhrusy@ubuntudesktop01:~$
```

```
fakhrusy@ubuntudesktop01:~$ ls /sys/devices/system/cpu/cpu0/cache/
index0 index1 index2 index3 uevent
fakhrusy@ubuntudesktop01:~$
```

```
fakhrusy@ubuntudesktop01:~$ for i in /sys/devices/system/cpu/cpu0/cache/index*; do
    echo "== $i ==";
    cat $i/level $i/type $i/size $i/coherency_line_size $i/shared_cpu_list;
done
== /sys/devices/system/cpu/cpu0/cache/index0 ==
1
Data
48K
64
0
== /sys/devices/system/cpu/cpu0/cache/index1 ==
1
Instruction
32K
64
0
== /sys/devices/system/cpu/cpu0/cache/index2 ==
2
Unified
1280K
64
0
== /sys/devices/system/cpu/cpu0/cache/index3 ==
3
Unified
24576K
64
0-1
fakhrusy@ubuntudesktop01:~$
```

Compare Physical Memory Information from Firmware and the Running Kernel

DIMM slots : 64
Module sizes : 8192 MB
Total installed memory : 1 (Ram socket #0)
Total memory seen by kernel : 7.7 Gi

Perbedaan installed dan usable memory karena Operating System dan hardware biasanya mengambil sebagian dari installed memory untuk menjalankan fungsi khusus, sehingga tidak dapat digunakan oleh user.

```
Handle 0x0222, DMI type 17, 34 bytes
Memory Device
  Array Handle: 0x0025
  Error Information Handle: No Error
  Total Width: 32 bits
  Data Width: 32 bits
  Size: No Module Installed
  Form Factor: DIMM
  Set: None
  Locator: NVD #63
  Bank Locator: NVD #63
  Type: Other
  Type Detail: Unknown
```

```
fakhrusy@ubuntudesktop01:~$
```

```
Handle 0x0085, DMI type 6, 12 bytes
Memory Module Information
  Socket Designation: RAM socket #0
  Bank Connections: None
  Current Speed: Unknown
  Type: EDO DIMM
  Installed Size: 8192 MB (Single-bank Connection)
  Enabled Size: 8192 MB (Single-bank Connection)
  Error Status: OK
```

```
fakhrusy@ubuntudesktop01:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        1.4Gi        4.9Gi        44Mi        1.8Gi        6.4Gi
Swap:          4.0Gi          0B        4.0Gi
fakhrusy@ubuntudesktop01:~$ \
```

Interpreting Linux Memory Utilization, Buffers, Cache, and Available Memory

- Used : Total memori aktif yang sedang digunakan oleh proses, kernel, dan sistem operasi.
- Free : RAM yang sedang tidak digunakan.
- Shared : Memori yang dapat diakses oleh beberapa program sekaligus.
- Buffers : Memori yang digunakan untuk menyangga data ketika proses transfer antar 2 lokasi.
- Cache : Memori yang digunakan oleh Operating System untuk menyimpan sementara sebuah file yang pernah dibaca dengan tujuan agar akses berikutnya dapat lebih cepat.
- Available : Estimasi kapasitas RAM yang sedang tidak digunakan dan available untuk menjalankan aplikasi baru tanpa swap

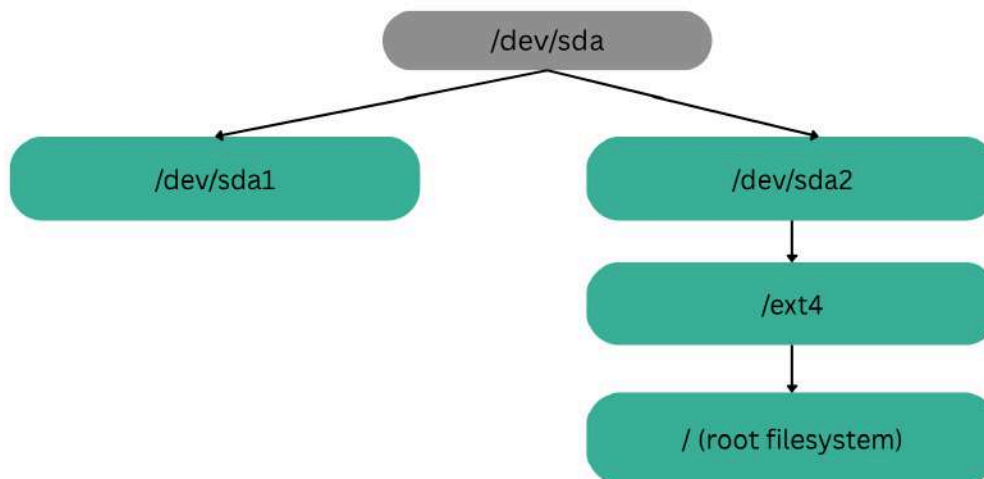
Sistem sedang menjalankan RAM dengan efisien.

```
fakhrusy@ubuntudesktop01:~$ free -h -w
              total        used        free      shared  buffers      cache   available
Mem:          7.7Gi        1.4Gi        4.8Gi        44Mi        50Mi        1.7Gi        6.3Gi
Swap:          4.0Gi          0B        4.0Gi
fakhrusy@ubuntudesktop01:~$ grep -E 'MemTotal|MemFree|MemAvailable|Buffers|Cached|SwapTotal|SwapFree' /proc/meminfo
MemTotal:      8083260 kB
MemFree:       5089072 kB
MemAvailable:  6671036 kB
Buffers:        51820 kB
Cached:        1764676 kB
SwapCached:      0 kB
SwapTotal:     4194300 kB
SwapFree:      4194300 kB
fakhrusy@ubuntudesktop01:~$ vmstat 1 5
procs -----memory----- --swap--  -----io----- -system--  -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi   bo    in   cs   us   sy   id   wa   st   gu
 1  0     0 5089072 51848 1828648    0    0    200    12  182    0    0    0 100    0    0    0
 0  0     0 5089072 51848 1828688    0    0     0     0  150  181    0    0 100    0    0    0
 0  0     0 5089072 51848 1828688    0    0     0     0  158  195    0    0 100    0    0    0
 0  0     0 5089072 51856 1828680    0    0     0    36  401  588    1    0 99    0    0    0
 0  0     0 5089072 51856 1828688    0    0     0     0  257  333    0    1 99    0    0    0
fakhrusy@ubuntudesktop01:~$
```


Map Storage Devices, Partitions, Filesystems, and Mount Points

Disk device names : sda
Partition names : sda1, sda2
Filesystem types : ext4
Mount points : /
Root filesystem device : /dev/sda2

lsblk menunjukkan relasi antara disk dan partisi, blkid untuk menampilkan informasi UUID dan filesystem. df -hT menunjukkan filesystem yang terpasang (mounted).



```
fakhrusy@buntudesktop01:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0    4K  1 loop /snap/bare/5
loop1       7:1      0   66.8M  1 loop /snap/core24/1587
loop2       7:2      0  251.7M  1 loop /snap/firefox/7766
loop3       7:3      0    74M  1 loop /snap/core22/2292
loop4       7:4      0  254.9M  1 loop /snap/firefox/7967
loop5       7:5      0   66.8M  1 loop /snap/core24/1499
loop6       7:6      0   18.5M  1 loop /snap/firmware-updater/212
loop7       7:7      0   16.4M  1 loop /snap/firmware-updater/216
loop8       7:8      0  531.4M  1 loop /snap/gnome-42-2204/247
loop9       7:9      0  606.1M  1 loop /snap/gnome-46-2404/153
loop10      7:10     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop11      7:11     0   395M  1 loop /snap/ mesa-2404/1165
loop12      7:12     0   10.8M  1 loop /snap/snap-store/1270
loop13      7:13     0    48.1M  1 loop /snap/snapd/25935
loop14      7:14     0    48.4M  1 loop /snap/snapd/26382
loop15      7:15     0  227.1M  1 loop /snap/thunderbird/1040
loop16      7:16     0  227.1M  1 loop /snap/thunderbird/1029
loop17      7:17     0    576K  1 loop /snap/snapd-desktop-integration/343
sda         8:0      0   40G  0 disk
├─sda1      8:1      0    1M  0 part
└─sda2      8:2      0   40G  0 part /
sr0        11:0     1    6.2G  0 rom  /media/fakhrusy/Ubuntu 24.04.4 LTS amd641
fakhrusy@buntudesktop01:~$
```

```
fakhrusy@ubuntudesktop01:~$ sudo blkid
[sudo] password for fakhrusy:
/dev/sda2: UUID="156a7993-b25b-4ce5-9742-3882711b4841" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="132cdb9c-19ca-443e-bbef-abc70a1e10f2"
/dev/loop1: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop17: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop8: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop15: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop6: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop13: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop4: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop11: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/sr0: BLOCK_SIZE="2048" UUID="2026-02-10-01-39-48-00" LABEL="Ubuntu 24.04.4 LTS amd64" TYPE="iso9660" PTTYPE="PMBR"
/dev/loop2: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop0: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop9: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop16: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop7: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/sda1: PARTUUID="e74c8f45-d0a8-453d-9012-9b5d85cf844d"
/dev/loop14: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop5: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop12: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop3: BLOCK_SIZE="131072" TYPE="squashfs"
/dev/loop10: BLOCK_SIZE="131072" TYPE="squashfs"
fakhrusy@ubuntudesktop01:~$
```

```
fakhrusy@ubuntudesktop01:~$ df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
tmpfs            tmpfs      790M  1.9M  788M   1% /run
/dev/sda2        ext4       40G   14G   24G  37% /
tmpfs            tmpfs      3.9G   0    3.9G   0% /dev/shm
tmpfs            tmpfs      5.0M   0    5.0M   0% /run/lock
tmpfs            tmpfs      790M  128K  790M   1% /run/user/1000
/dev/sr0         iso9660    6.2G   6.2G   0 100% /media/fakhrusy/Ubuntu 24.04.4 LTS amd641
fakhrusy@ubuntudesktop01:~$
```

Inspect PCI Devices and Kernel-Visible Bus Architecture

VGA/GPU : VMware SVGA II Adapter. Kernel driver vmwgfx

Ethernet controller : Intel Corporation 82545EM Gigabit Ethernet Controller (Copper) (rev 01)

SATA/NVME controller : VMware SATA AHCI controller

USB controller : VMware USB1.1 UHCI Controller, VMware USB2 EHCI Controller, VMware Device 077a

Audio device : Ensoniq ES1371/ES1373 / Creative Labs CT2518 (rev 02)

Sistem memiliki beberapa perangkat yang terhubung melalui pci, seperti GPU, Ethernet, SATA, USB controller dan juga audio device.

```
fakhrusy@ubuntudesktop01:~$ lspci
00:00.0 Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge (rev 01)
00:01.0 PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge (rev 01)
00:07.0 ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA (rev 08)
00:07.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:07.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:07.7 System peripheral: VMware Virtual Machine Communication Interface (rev 10)
00:0f.0 VGA compatible controller: VMware SVGA II Adapter
00:10.0 SCSI storage controller: Broadcom / LSI 53c1030 PCI-X Fusion-MPT Dual Ultra320 SCSI (rev 01)
00:11.0 PCI bridge: VMware PCI bridge (rev 02)
00:15.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.1 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.2 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.3 PCI bridge: VMware PCI Express Root Port (rev 01)

02:00.0 Ethernet controller: Intel Corporation 82545EM Gigabit Ethernet Controller (Copper) (rev 01)
02:01.0 SATA controller: VMware SATA AHCI controller
02:02.0 USB controller: VMware USB1.1 UHCI Controller
02:03.0 Multimedia audio controller: Ensoniq ES1371/ES1373 / Creative Labs CT2518 (rev 02)
02:04.0 USB controller: VMware USB2 EHCI Controller
03:00.0 USB controller: VMware Device 077a
```

```
fakhrusy@ubuntudesktop01:~$ lspci -k
00:00.0 Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge (rev 01)
Subsystem: VMware Virtual Machine Chipset
Kernel driver in use: agpgart-intel
00:01.0 PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge (rev 01)
00:07.0 ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA (rev 08)
Subsystem: VMware Virtual Machine Chipset
00:07.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
Subsystem: VMware Virtual Machine Chipset
Kernel driver in use: ata_piix
Kernel modules: pata_acpi
00:07.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
Subsystem: VMware Virtual Machine Chipset
Kernel modules: i2c_piix4
00:07.7 System peripheral: VMware Virtual Machine Communication Interface (rev 10)
Subsystem: VMware Virtual Machine Communication Interface
Kernel driver in use: vmw_vmci
Kernel modules: vmw_vmci
00:0f.0 VGA compatible controller: VMware SVGA II Adapter
Subsystem: VMware SVGA II Adapter
Kernel driver in use: vmwgfx
Kernel modules: vmwgfx
```


Examine Network Interface Hardware and Link Characteristics

Interface names : ens32

MAC address : 00:0c:29:b3:4e:7b

Driver : e1000

Link detected or not : yes

Speed : 1 Gbit/s

Duplex : full

Port type : twisted pair

NIC muncul di PCI dan juga Network Interface karena mewakili mewakili 2 hal yang berbeda. PCI adalah tempat dimana NIC dipasang di motherboard. Sedangkan network interface adalah antarmuka jaringan yang dibaca oleh OS.

```
fakhrusy@ubuntu01:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:b3:4e:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s0
fakhrusy@ubuntu01:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b3:4e:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.150.128/24 brd 192.168.150.255 scope global dynamic noprefixroute ens32
        valid_lft 1226sec preferred_lft 1226sec
    inet6 fe80::20c:29ff:feb3:4e7b/64 scope link
        valid_lft forever preferred_lft forever
```

```
fakhrusy@ubuntu01:~$ sudo lshw -class network
[sudo] password for fakhrusy:
*-network
   description: Ethernet interface
   product: 82545EM Gigabit Ethernet Controller (Copper)
   vendor: Intel Corporation
   physical id: 0
   bus info: pci@0000:02:00.0
   logical name: ens32
   version: 01
   serial: 00:0c:29:b3:4e:7b
   size: 1Gbit/s
   capacity: 1Gbit/s
   width: 64 bits
   clock: 66MHz
   capabilities: pm pcix bus_master cap_list rom ethernet physical logical tp 10bt 10bt-fd 100bt 100bt-fd 1000bt-fd autonegotia
   configuration: autonegotiation=on broadcast=yes driver=e1000 driverversion=6.17.0-19-generic duplex=full ip=192.168.150.128
   latency=0 link=yes mingnt=255 multicast=yes port=twisted pair speed=1Gbit/s
   resources: irq:18 memory:fd5c0000-fd5dffff memory:fdff0000-fdffff memory:fd500000-fd50ffff
```

Investigate NUMA, CPU Locality, and Resource Placement

Number of NUMA nodes : 1 node

CPUs in each node : 4

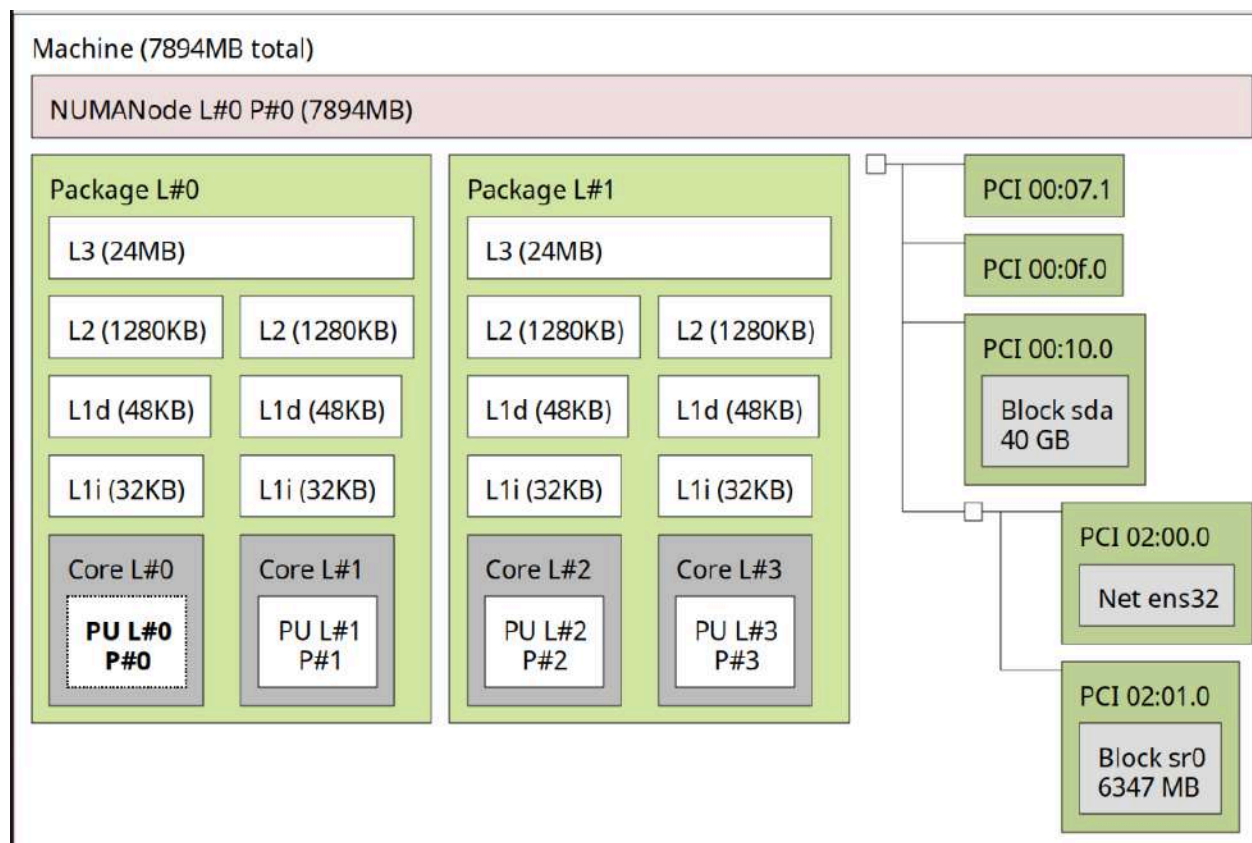
Memory size per node : Total memory 7893 MB

Termasuk ke dalam UMA-like karena hanya memiliki 1 node. Implikasinya adalah semua CPU mengakses satu memori utama yang sama sehingga latensi akan sama untuk semua core, akan tetapi ada kemungkinan bottleneck di memory bus saat banyak core bersaing.

Locality penting untuk beberapa hal. Database pada dasarnya bekerja dengan mengambil (fetch) baris dan buffer secepat mungkin. Dengan menempatkan data yang sering diakses secara berdekatan, sistem dapat beroperasi dengan lebih cepat.

Dalam virtualization, Virtual Machine harus mengakses cpu dan memory dengan cepat. Apabila seluruh resource saling berdekatan, maka performanya juga akan meningkat.

Untuk multi-thread workloads, locality penting untuk mengurangi latensi akses pada memori dan juga mengurangi bottleneck antar thread.




```

fakhrusy@ubuntudesktop01:~$ lscpu | grep -i numa
NUMA node(s):                  1
NUMA node0 CPU(s):             0-3
fakhrusy@ubuntudesktop01:~$ numactl --hardware
available: 1 nodes (0)
node 0 cpus: 0 1 2 3
node 0 size: 7893 MB
node 0 free: 4896 MB
node distances:
node   0
 0:    10
fakhrusy@ubuntudesktop01:~$ lstopo

```

Correlate Kernel Boot Messages with Detected Hardware

CPU Initialization

[0.000000] - [0.257528]	: Deteksi topologi CPU
[0.257553] - [0.381976]	: Per-CPU memory allocation
[0.491813] - [0.527578]	: inisialisasi kernel subsystem
[0.552563] - [0.554586]	: Identifikasi CPU0
[0.567683] - [0.574911]	: CPU sekunder
[0.595975] - [1.357667]	: Pengatur daya CPU
[9.843825] - [260.303907]	: Warning

```
fakhrusy@ubuntudesktop01:~$ sudo dmesg | grep -Ei 'cpu'
```

```
[ 0.000000] KERNEL supported cpus:
[ 0.257523] CPU topo: Max. logical packages: 64
[ 0.257524] CPU topo: Max. logical dies: 64
[ 0.257524] CPU topo: Max. dies per package: 1
[ 0.257527] CPU topo: Max. threads per core: 1
[ 0.257527] CPU topo: Num. cores per package: 2
[ 0.257528] CPU topo: Num. threads per package: 2
[ 0.257528] CPU topo: Allowing 4 present CPUs plus 124 hotplug CPUs
[ 0.257553] setup_percpu: NR_CPUS:8192 nr_cpumask_bits:128 nr_cpu_ids:128 nr_node_ids:1
[ 0.381804] percpu: Embedded 84 pages/cpu s221184 r8192 d114688 u524288
[ 0.381837] pcpu-alloc: s221184 r8192 d114688 u524288 alloc=1*2097152
[ 0.381839] pcpu-alloc: [0] 000 001 002 003 [0] 004 005 006 007
[ 0.381842] pcpu-alloc: [0] 008 009 010 011 [0] 012 013 014 015
[ 0.381845] pcpu-alloc: [0] 016 017 018 019 [0] 020 021 022 023
[ 0.381847] pcpu-alloc: [0] 024 025 026 027 [0] 028 029 030 031
[ 0.381850] pcpu-alloc: [0] 032 033 034 035 [0] 036 037 038 039
[ 0.381852] pcpu-alloc: [0] 040 041 042 043 [0] 044 045 046 047
[ 0.381855] pcpu-alloc: [0] 048 049 050 051 [0] 052 053 054 055
[ 0.381857] pcpu-alloc: [0] 056 057 058 059 [0] 060 061 062 063
[ 0.381860] pcpu-alloc: [0] 064 065 066 067 [0] 068 069 070 071
[ 0.381862] pcpu-alloc: [0] 072 073 074 075 [0] 076 077 078 079
[ 0.381865] pcpu-alloc: [0] 080 081 082 083 [0] 084 085 086 087
[ 0.381867] pcpu-alloc: [0] 088 089 090 091 [0] 092 093 094 095
[ 0.381870] pcpu-alloc: [0] 096 097 098 099 [0] 100 101 102 103
[ 0.381872] pcpu-alloc: [0] 104 105 106 107 [0] 108 109 110 111
[ 0.381875] pcpu-alloc: [0] 112 113 114 115 [0] 116 117 118 119
[ 0.381877] pcpu-alloc: [0] 120 121 122 123 [0] 124 125 126 127
[ 0.381975] printk: log_buf_len individual max cpu contribution: 4096 bytes
[ 0.381976] printk: log_buf_len total cpu_extra contributions: 520192 bytes
```

```
[ 0.381976] printk: log_buf_len total cpu_extra contributions: 520192 bytes
[ 0.491813] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=128, Nodes=1
[ 0.506240] rcu: RCU restricting CPUs from NR_CPUS=8192 to nr_cpu_ids=128.
[ 0.506243] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=128
[ 0.506557] RCU Tasks: Setting shift to 7 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=128.
[ 0.506563] RCU Tasks Rude: Setting shift to 7 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=128.
[ 0.506570] RCU Tasks Trace: Setting shift to 7 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=128.
[ 0.527012] x86/cpu: User Mode Instruction Prevention (UMIP) activated
[ 0.527578] MDS: Mitigation: Clear CPU buffers
[ 0.552563] smpboot: CPU0: 12th Gen Intel(R) Core(TM) i7-12700H (family: 0x6, model: 0x9a, stepping: 0x3)
[ 0.554563] core: CPUID marked event: 'cpu cycles' unavailable
[ 0.554563] core: CPUID marked event: 'instructions' unavailable
[ 0.554563] core: CPUID marked event: 'bus cycles' unavailable
[ 0.554563] core: CPUID marked event: 'cache references' unavailable
[ 0.554563] core: CPUID marked event: 'cache misses' unavailable
[ 0.554563] core: CPUID marked event: 'branch instructions' unavailable
[ 0.554563] core: CPUID marked event: 'branch misses' unavailable
[ 0.554568] core: cpu_core PMU driver:
[ 0.567683] smp: Bringing up secondary CPUs ...
[ 0.568827] .... node #0, CPUs: #1 #2 #3
[ 0.574911] smp: Brought up 1 node, 4 CPUs
[ 0.595975] cpuidle: using governor ladder
[ 0.595975] cpuidle: using governor menu
[ 1.275741] ledtrig-cpu: registered to indicate activity on CPUs
[ 1.357667] intel_pstate: CPU model not supported
[ 9.843825] WARNING: CPU: 3 PID: 316 at drivers/gpu/drm/vmwgfx/vmwgfx_bo.c:77 vmw_bo_free+0x41/0x60 [vmwgfx]
[ 9.843905] CPU: 3 UID: 0 PID: 316 Comm: plymouthd Not tainted 6.17.0-19-generic #19-24.04.2-Ubuntu PREEMPT(voluntary)
[ 260.062902] workqueue: netstamp_clear hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
[ 260.303907] workqueue: netstamp_clear hogged CPU for >10000us 5 times, consider switching to WQ_UNBOUND
```

Memory detection/reservation

[0.000000] : Mendeteksi slot fisik

[0.007906] - [0.007911] : Reservasi ACPI

[0.008615] : Pemetaan range fisik memori
 [0.257240] - [0.257543] : Registrasi nosave memory
 [0.549013] - [0.577830] : Reservasi memori untuk kernel dan firmware
 [1.530295] - [1.566066] : Pembebasan memori
 [4.077826] - [4.186046] : Memori runtime

```
fakhrusy@ubuntu:~$ sudo dmesg | grep -Ei 'memory'
[ 0.000000] DMI: Memory slots populated: 1/128
[ 0.007906] ACPI: Reserving FACP table memory at [mem 0xbfeef73-0xbfeef66]
[ 0.007907] ACPI: Reserving DSDT table memory at [mem 0xbfedd001-0xbfeef72]
[ 0.007908] ACPI: Reserving FACS table memory at [mem 0xbfeffffc0-0xbfefffff]
[ 0.007908] ACPI: Reserving FACS table memory at [mem 0xbfeffffc0-0xbfefffff]
[ 0.007909] ACPI: Reserving BOOT table memory at [mem 0xbfedcfb4-0xbfedcfdb]
[ 0.007909] ACPI: Reserving APIC table memory at [mem 0xbfedc872-0xbfedcfb3]
[ 0.007910] ACPI: Reserving MCFG table memory at [mem 0xbfedc836-0xbfedc871]
[ 0.007910] ACPI: Reserving SRAT table memory at [mem 0xbfedb72f-0xbfedbffe]
[ 0.007911] ACPI: Reserving HPET table memory at [mem 0xbfedb6f7-0xbfedb72e]
[ 0.007911] ACPI: Reserving WAET table memory at [mem 0xbfedb6cf-0xbfedb6f6]
[ 0.008615] Early memory node ranges
[ 0.257540] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x0000ffff]
[ 0.257541] PM: hibernation: Registered nosave memory: [mem 0x00009e000-0x0000ffff]
[ 0.257542] PM: hibernation: Registered nosave memory: [mem 0xbfed0000-0xbfefffff]
[ 0.257543] PM: hibernation: Registered nosave memory: [mem 0xc0000000-0xffffffff]
[ 0.549013] Freeing SMP alternatives memory: 52K
[ 0.576848] Memory: 7979760K/8388020K available (22213K kernel code, 4650K rwddata, 16292K rodata, 5268K init, 4212K bss, 388272K reserved, 0K cma-reserved)
[ 0.577830] x86/mm: Memory block size: 128MB
[ 1.530295] Freeing initrd memory: 75844K
[ 1.564394] Freeing unused decrypted memory: 2028K
[ 1.565384] Freeing unused kernel image (initmem) memory: 5268K
[ 1.565882] Freeing unused kernel image (text/rodata gap) memory: 312K
[ 1.566066] Freeing unused kernel image (rodata/data gap) memory: 92K
[ 4.077826] systemd[1]: Listening on systemd-oomd.socket - Userspace Out-Of-Memory (OOM) Killer Socket.
[ 4.186042] vmwgfx 0000:00:0f.0: [drm] Legacy memory limits: VRAM = 4096 KiB, FIFO = 256 KiB, surface = 0 KiB
[ 4.186046] vmwgfx 0000:00:0f.0: [drm] Maximum display memory size is 262144 KiB
```

PCI enumeration

[0.007876]-[0.281245] : Persiapan awal PCI
 [0.657698]-[0.700260] : Inisialisasi subsistem PCI
 [0.856276]-[0.858504] : Root bridge & bus resource
 [0.858678]-[0.881668] : bus 00
 [0.885694]-[0.925038] : root ports
 [0.933922]-[1.016187] : bus sekunder
 [1.065240]-[1.272392] : irq & vga
 [1.332541]-[1.335720] : alokasi resource
 [1.338807]-[1.411870] : finalisasi


```
fakhrusy@ubuntu:~$ sudo dmesg | grep -Ei 'pci'
[sudo] password for fakhrusy:
[ 0.007876] ACPI: MCFG 0x00000000BFEDC836 00003C (v01 PILTD $PCITBL$ 06040000 LTP 00000001)
[ 0.281245] [mem 0xc0000000-0xffffffff] available for PCI devices
[ 0.657698] acpihp: ACPI Hot Plug PCI Controller Driver version: 0.5
[ 0.657819] PCI: ECAM [mem 0xf0000000-0xf7ffffff] (base 0xf0000000) for domain 0000 [bus 00-7f]
[ 0.657831] PCI: Using configuration type 1 for base access
[ 0.700259] PCI: Using host bridge windows from ACPI; if necessary, use "pci=nocrs" and report a bug
[ 0.700260] PCI: Using E820 reservations for host bridge windows
[ 0.856276] ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-7f])
[ 0.856533] acpi PNP0A03:00: _OSC: OS now controls [PCIeHotplug SHPCHotplug PME PCIECapability]
[ 0.858495] PCI host bridge to bus 0000:00
[ 0.858498] pci_bus 0000:00: root bus resource [mem 0x000a0000-0x000bffff window]
[ 0.858500] pci_bus 0000:00: root bus resource [mem 0x000d0000-0x000dbfff window]
[ 0.858501] pci_bus 0000:00: root bus resource [mem 0xc0000000-0xfebffffff window]
[ 0.858502] pci_bus 0000:00: root bus resource [io 0x0000-0x0cf7 window]
[ 0.858503] pci_bus 0000:00: root bus resource [io 0x0d00-0xffff window]
[ 0.858504] pci_bus 0000:00: root bus resource [bus 00-7f]
[ 0.858678] pci 0000:00:00.0: [8086:7190] type 00 class 0x060000 conventional PCI endpoint
[ 0.859387] pci 0000:00:01.0: [8086:7191] type 01 class 0x060400 conventional PCI bridge
[ 0.859605] pci 0000:00:01.0: PCI bridge to [bus 01]
[ 0.865885] pci 0000:00:07.0: [8086:7110] type 00 class 0x060100 conventional PCI endpoint
[ 0.866523] pci 0000:00:07.1: [8086:7111] type 00 class 0x060100 conventional PCI endpoint
[ 0.867205] pci 0000:00:07.1: BAR 4 [io 0x1060-0x106f]
[ 0.867248] pci 0000:00:07.1: BAR 0 [io 0x01f0-0x01f7]: legacy IDE quirk
[ 0.867249] pci 0000:00:07.1: BAR 1 [io 0x03f6]: legacy IDE quirk
[ 0.867250] pci 0000:00:07.1: BAR 2 [io 0x0170-0x0177]: legacy IDE quirk
[ 0.867250] pci 0000:00:07.1: BAR 3 [io 0x0376]: legacy IDE quirk
[ 0.867463] pci 0000:00:07.3: [8086:7113] type 00 class 0x068000 conventional PCI endpoint
[ 0.868064] pci 0000:00:07.3: quirk: [io 0x1000-0x103f] claimed by PIIX4 ACPI
[ 0.868074] pci 0000:00:07.3: quirk: [io 0x1040-0x104f] claimed by PIIX4 SMB
[ 0.868592] pci 0000:00:07.7: [15ad:0740] type 00 class 0x088000 conventional PCI endpoint
```

```
[ 0.880844] pci 0000:00:10.0: ROM [mem 0x00000000-0x00003fff pref]
[ 0.881359] pci 0000:00:11.0: [15ad:0790] type 01 class 0x060401 conventional PCI bridge
[ 0.881556] pci 0000:00:11.0: PCI bridge to [bus 02] (subtractive decode)
[ 0.881575] pci 0000:00:11.0: bridge window [io 0x2000-0x3fff]
[ 0.881593] pci 0000:00:11.0: bridge window [mem 0xfd500000-0xfdffffff]
[ 0.881668] pci 0000:00:11.0: bridge window [mem 0xe7b00000-0xe7ffffff 64bit pref]
[ 0.885694] pci 0000:00:15.0: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.885901] pci 0000:00:15.0: PCI bridge to [bus 03]
[ 0.885938] pci 0000:00:15.0: bridge window [mem 0xfd400000-0xfd4ffffff]
[ 0.886457] pci 0000:00:15.0: PME# supported from D0 D3hot D3cold
[ 0.886876] pci 0000:00:15.1: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.887078] pci 0000:00:15.1: PCI bridge to [bus 04]
[ 0.887097] pci 0000:00:15.1: bridge window [io 0x7000-0x7fff]
[ 0.887115] pci 0000:00:15.1: bridge window [mem 0xfd000000-0xfd0ffffff]
[ 0.887192] pci 0000:00:15.1: bridge window [mem 0xe7700000-0xe77ffffff 64bit pref]
[ 0.887647] pci 0000:00:15.1: PME# supported from D0 D3hot D3cold
[ 0.888069] pci 0000:00:15.2: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.888302] pci 0000:00:15.2: PCI bridge to [bus 05]
[ 0.888327] pci 0000:00:15.2: bridge window [io 0xb000-0xbfff]
[ 0.888350] pci 0000:00:15.2: bridge window [mem 0xfcc00000-0xfccffffff]
[ 0.888428] pci 0000:00:15.2: bridge window [mem 0xe7300000-0xe73ffffff 64bit pref]
[ 0.888926] pci 0000:00:15.2: PME# supported from D0 D3hot D3cold
[ 0.889414] pci 0000:00:15.3: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.889629] pci 0000:00:15.3: PCI bridge to [bus 06]
[ 0.889666] pci 0000:00:15.3: bridge window [mem 0xfc800000-0xfc8ffffff]
[ 0.889745] pci 0000:00:15.3: bridge window [mem 0xe6f00000-0xe6ffffff 64bit pref]
[ 0.890182] pci 0000:00:15.3: PME# supported from D0 D3hot D3cold
[ 0.890567] pci 0000:00:15.4: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.890773] pci 0000:00:15.4: PCI bridge to [bus 07]
[ 0.890809] pci 0000:00:15.4: bridge window [mem 0xfc400000-0xfc4ffffff]
[ 0.890886] pci 0000:00:15.4: bridge window [mem 0xe6b00000-0xe6bffffff 64bit pref]
[ 0.891315] pci 0000:00:15.4: PME# supported from D0 D3hot D3cold
```



```
[ 0.891749] pci 0000:00:15.5: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.891948] pci 0000:00:15.5: PCI bridge to [bus 08]
[ 0.891983] pci 0000:00:15.5: bridge window [mem 0xfcb00000-0xfcbffffff]
[ 0.892059] pci 0000:00:15.5: bridge window [mem 0xe6700000-0xe67ffffff 64bit pref]
[ 0.892494] pci 0000:00:15.5: PME# supported from D0 D3hot D3cold
[ 0.892883] pci 0000:00:15.6: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.893082] pci 0000:00:15.6: PCI bridge to [bus 09]
[ 0.893118] pci 0000:00:15.6: bridge window [mem 0xfbc00000-0xfbcffffff]
[ 0.893194] pci 0000:00:15.6: bridge window [mem 0xe6300000-0xe63ffffff 64bit pref]
[ 0.893630] pci 0000:00:15.6: PME# supported from D0 D3hot D3cold
[ 0.894076] pci 0000:00:15.7: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.894273] pci 0000:00:15.7: PCI bridge to [bus 0a]
[ 0.894308] pci 0000:00:15.7: bridge window [mem 0xfbb00000-0xfbbffffff]
[ 0.894392] pci 0000:00:15.7: bridge window [mem 0xe5f00000-0xe5fffffff 64bit pref]
[ 0.894833] pci 0000:00:15.7: PME# supported from D0 D3hot D3cold
[ 0.895220] pci 0000:00:16.0: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.895426] pci 0000:00:16.0: PCI bridge to [bus 0b]
[ 0.895446] pci 0000:00:16.0: bridge window [io 0x4000-0x4fff]
[ 0.895465] pci 0000:00:16.0: bridge window [mem 0xfd300000-0xfd3ffffff]
[ 0.895548] pci 0000:00:16.0: bridge window [mem 0xe7a00000-0xe7affffff 64bit pref]
[ 0.895992] pci 0000:00:16.0: PME# supported from D0 D3hot D3cold
[ 0.896381] pci 0000:00:16.1: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.896586] pci 0000:00:16.1: PCI bridge to [bus 0c]
[ 0.896612] pci 0000:00:16.1: bridge window [io 0x8000-0x8fff]
[ 0.896638] pci 0000:00:16.1: bridge window [mem 0xfcf00000-0xfcfffffff]
[ 0.896709] pci 0000:00:16.1: bridge window [mem 0xe7600000-0xe76ffffff 64bit pref]
[ 0.897149] pci 0000:00:16.1: PME# supported from D0 D3hot D3cold
[ 0.897575] pci 0000:00:16.2: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.897787] pci 0000:00:16.2: PCI bridge to [bus 0d]
[ 0.897807] pci 0000:00:16.2: bridge window [io 0xc000-0xcfff]
[ 0.897825] pci 0000:00:16.2: bridge window [mem 0xfcb00000-0xfcbffffff]
[ 0.897903] pci 0000:00:16.2: bridge window [mem 0xe7200000-0xe72ffffff 64bit pref]
```

```
[ 0.899172] pci 0000:00:16.3: bridge window [mem 0xe6e00000-0xe6effffff 64bit pref]
[ 0.900362] pci 0000:00:16.3: PME# supported from D0 D3hot D3cold
[ 0.900823] pci 0000:00:16.4: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.901029] pci 0000:00:16.4: PCI bridge to [bus 0f]
[ 0.901067] pci 0000:00:16.4: bridge window [mem 0xfcb00000-0xfcbffffff]
[ 0.901145] pci 0000:00:16.4: bridge window [mem 0xe6a00000-0xe6affffff 64bit pref]
[ 0.901624] pci 0000:00:16.4: PME# supported from D0 D3hot D3cold
[ 0.902046] pci 0000:00:16.5: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.902248] pci 0000:00:16.5: PCI bridge to [bus 10]
[ 0.902284] pci 0000:00:16.5: bridge window [mem 0xfbf00000-0xfbfffffff]
[ 0.902384] pci 0000:00:16.5: bridge window [mem 0xe6600000-0xe66ffffff 64bit pref]
[ 0.902816] pci 0000:00:16.5: PME# supported from D0 D3hot D3cold
[ 0.903266] pci 0000:00:16.6: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.903472] pci 0000:00:16.6: PCI bridge to [bus 11]
[ 0.903508] pci 0000:00:16.6: bridge window [mem 0xfbb00000-0xfbbffffff]
[ 0.903582] pci 0000:00:16.6: bridge window [mem 0xe6200000-0xe62ffffff 64bit pref]
[ 0.904069] pci 0000:00:16.6: PME# supported from D0 D3hot D3cold
[ 0.904659] pci 0000:00:16.7: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.904870] pci 0000:00:16.7: PCI bridge to [bus 12]
[ 0.904909] pci 0000:00:16.7: bridge window [mem 0xfbb00000-0xfbbffffff]
[ 0.904990] pci 0000:00:16.7: bridge window [mem 0xe5e00000-0xe5effffff 64bit pref]
[ 0.905432] pci 0000:00:16.7: PME# supported from D0 D3hot D3cold
[ 0.905817] pci 0000:00:17.0: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.906009] pci 0000:00:17.0: PCI bridge to [bus 13]
[ 0.906028] pci 0000:00:17.0: bridge window [io 0x5000-0x5fff]
[ 0.906045] pci 0000:00:17.0: bridge window [mem 0xfd200000-0xfd2ffffff]
[ 0.906119] pci 0000:00:17.0: bridge window [mem 0xe7900000-0xe79ffffff 64bit pref]
[ 0.906537] pci 0000:00:17.0: PME# supported from D0 D3hot D3cold
[ 0.906957] pci 0000:00:17.1: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.907149] pci 0000:00:17.1: PCI bridge to [bus 14]
[ 0.907167] pci 0000:00:17.1: bridge window [io 0x9000-0x9fff]
[ 0.907185] pci 0000:00:17.1: bridge window [mem 0xfce00000-0xfceffffff]
```

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[ 0.910834] pci 0000:00:17.4: bridge window [mem 0xe6900000-0xe69fffff 64bit pref]
[ 0.911269] pci 0000:00:17.4: PME# supported from D0 D3hot D3cold
[ 0.911652] pci 0000:00:17.5: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.911861] pci 0000:00:17.5: PCI bridge to [bus 18]
[ 0.911897] pci 0000:00:17.5: bridge window [mem 0xfbe00000-0xfbefffff]
[ 0.911973] pci 0000:00:17.5: bridge window [mem 0xe6500000-0xe65fffff 64bit pref]
[ 0.912406] pci 0000:00:17.5: PME# supported from D0 D3hot D3cold
[ 0.912785] pci 0000:00:17.6: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.912984] pci 0000:00:17.6: PCI bridge to [bus 19]
[ 0.913019] pci 0000:00:17.6: bridge window [mem 0xfba00000-0xfbafffff]
[ 0.913096] pci 0000:00:17.6: bridge window [mem 0xe6100000-0xe61fffff 64bit pref]
[ 0.913524] pci 0000:00:17.6: PME# supported from D0 D3hot D3cold
[ 0.914008] pci 0000:00:17.7: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.914206] pci 0000:00:17.7: PCI bridge to [bus 1a]
[ 0.914246] pci 0000:00:17.7: bridge window [mem 0xfba00000-0xfbafffff]
[ 0.914322] pci 0000:00:17.7: bridge window [mem 0xe5d00000-0xe5dfffff 64bit pref]
[ 0.914765] pci 0000:00:17.7: PME# supported from D0 D3hot D3cold
[ 0.915412] pci 0000:00:18.0: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.915682] pci 0000:00:18.0: PCI bridge to [bus 1b]
[ 0.915705] pci 0000:00:18.0: bridge window [io 0x6000-0x6fff]
[ 0.915725] pci 0000:00:18.0: bridge window [mem 0xfcd10000-0xfcd1fffff]
[ 0.915808] pci 0000:00:18.0: bridge window [mem 0xe7800000-0xe78fffff 64bit pref]
[ 0.916251] pci 0000:00:18.0: PME# supported from D0 D3hot D3cold
[ 0.916699] pci 0000:00:18.1: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.916901] pci 0000:00:18.1: PCI bridge to [bus 1c]
[ 0.916920] pci 0000:00:18.1: bridge window [io 0xa000-0xafff]
[ 0.916938] pci 0000:00:18.1: bridge window [mem 0xfcd00000-0xfcdfffff]
[ 0.917200] pci 0000:00:18.1: bridge window [mem 0xe7400000-0xe74fffff 64bit pref]
[ 0.917599] pci 0000:00:18.1: PME# supported from D0 D3hot D3cold
[ 0.918104] pci 0000:00:18.2: [15ad:07a0] type 01 class 0x060400 PCIe Root Port
[ 0.918396] pci 0000:00:18.2: PCI bridge to [bus 1d]
[ 0.918425] pci 0000:00:18.2: bridge window [io 0xe000-0xefff]
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[ 0.924609] pci 0000:00:18.7: bridge window [mem 0xe5c00000-0xe5cfffff 64bit pref]
[ 0.925038] pci 0000:00:18.7: PME# supported from D0 D3hot D3cold
[ 0.933922] pci_bus 0000:01: extended config space not accessible
[ 0.973647] pci 0000:00:01.0: PCI bridge to [bus 01]
[ 0.973837] pci_bus 0000:02: extended config space not accessible
[ 0.974657] pci 0000:02:00.0: [8086:100f] type 00 class 0x020000 conventional PCI endpoint
[ 0.975368] pci 0000:02:00.0: BAR 0 [mem 0xfdc00000-0xfdc5ffff 64bit]
[ 0.975387] pci 0000:02:00.0: BAR 2 [mem 0xfdf00000-0xfdf5ffff 64bit]
[ 0.975397] pci 0000:02:00.0: BAR 4 [io 0x2000-0x203f]
[ 0.975415] pci 0000:02:00.0: ROM [mem 0xe0000000-0xe000ffff pref]
[ 0.975668] pci 0000:02:00.0: PME# supported from D0 D3hot D3cold
[ 0.976206] pci 0000:02:01.0: [15ad:07e0] type 00 class 0x010601 conventional PCI endpoint
[ 0.976907] pci 0000:02:01.0: BAR 5 [mem 0xfdf00000-0xfdf5ffff]
[ 0.976918] pci 0000:02:01.0: ROM [mem 0xe0000000-0xe000ffff pref]
[ 0.977371] pci 0000:02:01.0: PME# supported from D3hot
[ 0.977878] pci 0000:02:02.0: [15ad:0774] type 00 class 0x0c0300 conventional PCI endpoint
[ 0.978697] pci 0000:02:02.0: BAR 4 [io 0x2080-0x209f]
[ 0.979585] pci 0000:02:03.0: [1274:1371] type 00 class 0x040100 conventional PCI endpoint
[ 0.980239] pci 0000:02:03.0: BAR 0 [io 0x2040-0x207f]
[ 0.980824] pci 0000:02:04.0: [15ad:0779] type 00 class 0x0c0320 conventional PCI endpoint
[ 0.981359] pci 0000:02:04.0: BAR 0 [mem 0xfdf50000-0xfdf5ffff]
[ 1.013930] pci 0000:00:11.0: PCI bridge to [bus 02] (subtractive decode)
[ 1.014002] pci 0000:00:11.0: bridge window [mem 0x000a0000-0x000bffff window] (subtractive decode)
[ 1.014004] pci 0000:00:11.0: bridge window [mem 0x000d0000-0x000dbfff window] (subtractive decode)
[ 1.014005] pci 0000:00:11.0: bridge window [mem 0xc0000000-0xc000ffff window] (subtractive decode)
[ 1.014006] pci 0000:00:11.0: bridge window [io 0x0000-0x0cf7 window] (subtractive decode)
[ 1.014007] pci 0000:00:11.0: bridge window [io 0x0d00-0x0fff window] (subtractive decode)
[ 1.014451] pci 0000:03:00.0: [15ad:077a] type 00 class 0x0c0330 PCIe Endpoint
[ 1.015098] pci 0000:03:00.0: BAR 0 [mem 0xfdf40000-0xfdf4ffff 64bit]
[ 1.015563] pci 0000:03:00.0: PME# supported from D0 D3hot D3cold
[ 1.016187] pci 0000:03:00.0: disabling ASPM on pre-1.1 PCIe device. You can enable it with 'pcie_aspm=force'
[ 1.016284] pci 0000:00:15.0: PCI bridge to [bus 03]
[ 1.017719] pci 0000:00:15.1: PCI bridge to [bus 04]
[ 1.019152] pci 0000:00:15.2: PCI bridge to [bus 05]
[ 1.020591] pci 0000:00:15.3: PCI bridge to [bus 06]
[ 1.022024] pci 0000:00:15.4: PCI bridge to [bus 07]
[ 1.023455] pci 0000:00:15.5: PCI bridge to [bus 08]
[ 1.025257] pci 0000:00:15.6: PCI bridge to [bus 09]
```



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[ 1.060012] pci 0000:00:18.5: PCI bridge to [bus 20]
[ 1.061702] pci 0000:00:18.6: PCI bridge to [bus 21]
[ 1.063259] pci 0000:00:18.7: PCI bridge to [bus 22]
[ 1.065240] ACPI: PCI: Interrupt link LNKA configured for IRQ 9
[ 1.065281] ACPI: PCI: Interrupt link LNKB configured for IRQ 11
[ 1.065333] ACPI: PCI: Interrupt link LNKC configured for IRQ 10
[ 1.065340] ACPI: PCI: Interrupt link LNKD configured for IRQ 7
[ 1.094415] PCI: Using ACPI for IRQ routing
[ 1.270691] PCI: pci_cache_line_size set to 64 bytes
[ 1.272388] pci 0000:00:0f.0: vgaarb: setting as boot VGA device
[ 1.272391] pci 0000:00:0f.0: vgaarb: bridge control possible
[ 1.272392] pci 0000:00:0f.0: vgaarb: VGA device added: decodes=io+mem,owns=io+mem,locks=None
[ 1.332541] pci 0000:00:15.0: bridge window [io 0x1000-0x0fff] to [bus 03] add_size 1000
[ 1.332545] pci 0000:00:15.0: bridge window [mem 0x00100000-0x000fffff 64bit pref] to [bus 03] add_size 200000 add_align 100000
[ 1.332547] pci 0000:00:15.3: bridge window [io 0x1000-0x0fff] to [bus 06] add_size 1000
[ 1.332548] pci 0000:00:15.4: bridge window [io 0x1000-0x0fff] to [bus 07] add_size 1000
[ 1.332549] pci 0000:00:15.5: bridge window [io 0x1000-0x0fff] to [bus 08] add_size 1000
[ 1.332550] pci 0000:00:15.6: bridge window [io 0x1000-0x0fff] to [bus 09] add_size 1000
[ 1.332551] pci 0000:00:15.7: bridge window [io 0x1000-0x0fff] to [bus 0a] add_size 1000
[ 1.332552] pci 0000:00:16.3: bridge window [io 0x1000-0x0fff] to [bus 0e] add_size 1000
[ 1.332553] pci 0000:00:16.4: bridge window [io 0x1000-0x0fff] to [bus 0f] add_size 1000
[ 1.332554] pci 0000:00:16.5: bridge window [io 0x1000-0x0fff] to [bus 10] add_size 1000
[ 1.332555] pci 0000:00:16.6: bridge window [io 0x1000-0x0fff] to [bus 11] add_size 1000
[ 1.332556] pci 0000:00:16.7: bridge window [io 0x1000-0x0fff] to [bus 12] add_size 1000
[ 1.332557] pci 0000:00:17.3: bridge window [io 0x1000-0x0fff] to [bus 16] add_size 1000
[ 1.332558] pci 0000:00:17.4: bridge window [io 0x1000-0x0fff] to [bus 17] add_size 1000
[ 1.332559] pci 0000:00:17.5: bridge window [io 0x1000-0x0fff] to [bus 18] add_size 1000
[ 1.332560] pci 0000:00:17.6: bridge window [io 0x1000-0x0fff] to [bus 19] add_size 1000
[ 1.332561] pci 0000:00:17.7: bridge window [io 0x1000-0x0fff] to [bus 1a] add_size 1000
[ 1.332562] pci 0000:00:18.3: bridge window [io 0x1000-0x0fff] to [bus 1e] add_size 1000
[ 1.332563] pci 0000:00:18.4: bridge window [io 0x1000-0x0fff] to [bus 1f] add_size 1000
[ 1.332564] pci 0000:00:18.5: bridge window [io 0x1000-0x0fff] to [bus 20] add_size 1000
[ 1.332565] pci 0000:00:18.6: bridge window [io 0x1000-0x0fff] to [bus 21] add_size 1000
[ 1.332566] pci 0000:00:18.7: bridge window [io 0x1000-0x0fff] to [bus 22] add_size 1000
[ 1.332566] pci 0000:00:15.0: bridge window [mem 0xc0000000-0xc01fffff 64bit pref]: assigned
[ 1.332588] pci 0000:00:10.0: ROM [mem 0xc0200000-0xc0203fff pref]: assigned
[ 1.332590] pci 0000:00:15.0: bridge window [io size 0x1000]: can't assign; no space
[ 1.332592] pci 0000:00:15.0: bridge window [io size 0x1000]: failed to assign
```

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[ 1.332598] pci 0000:00:15.6: bridge window [io size 0x1000]: can't assign; no space
[ 1.332598] pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
[ 1.332599] pci 0000:00:15.7: bridge window [io size 0x1000]: can't assign; no space
[ 1.332600] pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
[ 1.332601] pci 0000:00:16.3: bridge window [io size 0x1000]: can't assign; no space
[ 1.332601] pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
[ 1.332602] pci 0000:00:16.4: bridge window [io size 0x1000]: can't assign; no space
[ 1.332603] pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
[ 1.332604] pci 0000:00:16.5: bridge window [io size 0x1000]: can't assign; no space
[ 1.332604] pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
[ 1.332605] pci 0000:00:16.6: bridge window [io size 0x1000]: can't assign; no space
[ 1.332606] pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
[ 1.332607] pci 0000:00:16.7: bridge window [io size 0x1000]: can't assign; no space
[ 1.332608] pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
[ 1.332609] pci 0000:00:17.3: bridge window [io size 0x1000]: can't assign; no space
[ 1.332609] pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
[ 1.332610] pci 0000:00:17.4: bridge window [io size 0x1000]: can't assign; no space
[ 1.332611] pci 0000:00:17.4: bridge window [io size 0x1000]: failed to assign
[ 1.332612] pci 0000:00:17.5: bridge window [io size 0x1000]: can't assign; no space
[ 1.332612] pci 0000:00:17.5: bridge window [io size 0x1000]: failed to assign
[ 1.332613] pci 0000:00:17.6: bridge window [io size 0x1000]: can't assign; no space
[ 1.332614] pci 0000:00:17.6: bridge window [io size 0x1000]: failed to assign
[ 1.332615] pci 0000:00:17.7: bridge window [io size 0x1000]: can't assign; no space
[ 1.332615] pci 0000:00:17.7: bridge window [io size 0x1000]: failed to assign
[ 1.332616] pci 0000:00:18.3: bridge window [io size 0x1000]: can't assign; no space
[ 1.332617] pci 0000:00:18.3: bridge window [io size 0x1000]: failed to assign
[ 1.332618] pci 0000:00:18.4: bridge window [io size 0x1000]: can't assign; no space
[ 1.332619] pci 0000:00:18.4: bridge window [io size 0x1000]: failed to assign
[ 1.332619] pci 0000:00:18.5: bridge window [io size 0x1000]: can't assign; no space
[ 1.332620] pci 0000:00:18.5: bridge window [io size 0x1000]: failed to assign
[ 1.332621] pci 0000:00:18.6: bridge window [io size 0x1000]: can't assign; no space
[ 1.332622] pci 0000:00:18.6: bridge window [io size 0x1000]: failed to assign
[ 1.332623] pci 0000:00:18.7: bridge window [io size 0x1000]: can't assign; no space
[ 1.332623] pci 0000:00:18.7: bridge window [io size 0x1000]: failed to assign
[ 1.332626] pci 0000:00:18.7: bridge window [io size 0x1000]: can't assign; no space
[ 1.332627] pci 0000:00:18.7: bridge window [io size 0x1000]: failed to assign
[ 1.332628] pci 0000:00:18.6: bridge window [io size 0x1000]: can't assign; no space
[ 1.332628] pci 0000:00:18.6: bridge window [io size 0x1000]: failed to assign
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1.332668] pci 0000:00:01.0: PCI bridge to [bus 01]
1.332760] pci 0000:02:00.0: ROM [mem 0xfd500000-0xfd50ffff pref]: assigned
1.332761] pci 0000:02:01.0: ROM [mem 0xfd510000-0xfd51ffff pref]: assigned
1.332762] pci 0000:00:11.0: PCI bridge to [bus 02]
1.332772] pci 0000:00:11.0: bridge window [io 0x2000-0x3fff]
1.332796] pci 0000:00:11.0: bridge window [mem 0xfd500000-0xfdffffff]
1.332813] pci 0000:00:11.0: bridge window [mem 0xe7b00000-0xe7ffffff 64bit pref]
1.332845] pci 0000:00:15.0: PCI bridge to [bus 03]
1.332869] pci 0000:00:15.0: bridge window [mem 0xfd400000-0xfd4fffff]
1.332885] pci 0000:00:15.0: bridge window [mem 0xc0000000-0xc0ffffff 64bit pref]
1.332916] pci 0000:00:15.1: PCI bridge to [bus 04]
1.332925] pci 0000:00:15.1: bridge window [io 0x7000-0x7fff]
1.332949] pci 0000:00:15.1: bridge window [mem 0xfd000000-0xfd0fffff]
1.332965] pci 0000:00:15.1: bridge window [mem 0xe7700000-0xe77fffff 64bit pref]
1.332996] pci 0000:00:15.2: PCI bridge to [bus 05]
1.333005] pci 0000:00:15.2: bridge window [io 0xb000-0xbfff]
1.333029] pci 0000:00:15.2: bridge window [mem 0xfcc00000-0xfccfffff]
1.333045] pci 0000:00:15.2: bridge window [mem 0xe7300000-0xe73fffff 64bit pref]
1.333076] pci 0000:00:15.3: PCI bridge to [bus 06]
1.333100] pci 0000:00:15.3: bridge window [mem 0xfc000000-0xfc0fffff]
1.333116] pci 0000:00:15.3: bridge window [mem 0xe6f00000-0xe6ffffff 64bit pref]
1.333148] pci 0000:00:15.4: PCI bridge to [bus 07]
1.333179] pci 0000:00:15.4: bridge window [mem 0xfc000000-0xfc4fffff]
1.333195] pci 0000:00:15.4: bridge window [mem 0xe6b00000-0xe6bfffff 64bit pref]
1.333228] pci 0000:00:15.5: PCI bridge to [bus 08]
1.333252] pci 0000:00:15.5: bridge window [mem 0xfc000000-0xfc0fffff]
1.333268] pci 0000:00:15.5: bridge window [mem 0xe6700000-0xe67fffff 64bit pref]
1.333301] pci 0000:00:15.6: PCI bridge to [bus 09]
1.333326] pci 0000:00:15.6: bridge window [mem 0xfbc00000-0xfbcfffff]
1.333350] pci 0000:00:15.6: bridge window [mem 0xe6300000-0xe63fffff 64bit pref]
1.333384] pci 0000:00:15.7: PCI bridge to [bus 0a]
1.333590] pci 0000:00:15.7: bridge window [mem 0xfb000000-0xfb0fffff]
1.333607] pci 0000:00:15.7: bridge window [mem 0xe5f00000-0xe5ffffff 64bit pref]
1.333641] pci 0000:00:16.0: PCI bridge to [bus 0b]
1.333651] pci 0000:00:16.0: bridge window [io 0x4000-0x4fff]
1.333676] pci 0000:00:16.0: bridge window [mem 0xfd300000-0xfd3fffff]
1.333694] pci 0000:00:16.0: bridge window [mem 0xe7a00000-0xe7afffff 64bit pref]
1.333728] pci 0000:00:16.1: PCI bridge to [bus 0c]

1.335597] pci 0000:00:18.7: bridge window [mem 0xfb500000-0xfb5fffff]
1.335615] pci 0000:00:18.7: bridge window [mem 0xe5c00000-0xe5cfffff 64bit pref]
1.335651] pci_bus 0000:00: Some PCI device resources are unassigned, try booting with pci=realloc
1.335653] pci_bus 0000:00: resource 4 [mem 0x000a0000-0x000bffff window]
1.335654] pci_bus 0000:00: resource 5 [mem 0x000d0000-0x000dbfff window]
1.335655] pci_bus 0000:00: resource 6 [mem 0xc0000000-0xfebfffff window]
1.335656] pci_bus 0000:00: resource 7 [io 0x0000-0x0cf7 window]
1.335657] pci_bus 0000:00: resource 8 [io 0x0d00-0xfeff window]
1.335658] pci_bus 0000:02: resource 0 [io 0x2000-0x3fff]
1.335659] pci_bus 0000:02: resource 1 [mem 0xfd500000-0xfdffffff]
1.335660] pci_bus 0000:02: resource 2 [mem 0xe7b00000-0xe7ffffff 64bit pref]
1.335660] pci_bus 0000:02: resource 4 [mem 0x000a0000-0x000bffff window]
1.335661] pci_bus 0000:02: resource 5 [mem 0x000d0000-0x000dbfff window]
1.335662] pci_bus 0000:02: resource 6 [mem 0xc0000000-0xfebfffff window]
1.335663] pci_bus 0000:02: resource 7 [io 0x0000-0x0cf7 window]
1.335663] pci_bus 0000:02: resource 8 [io 0x0d00-0xfeff window]
1.335665] pci_bus 0000:03: resource 1 [mem 0xfd400000-0xfd4fffff]
1.335665] pci_bus 0000:03: resource 2 [mem 0xc0000000-0xc0ffffff 64bit pref]
1.335666] pci_bus 0000:04: resource 0 [io 0x7000-0x7fff]
1.335667] pci_bus 0000:04: resource 1 [mem 0xfd000000-0xfd0fffff]
1.335668] pci_bus 0000:04: resource 2 [mem 0xe7700000-0xe77fffff 64bit pref]
1.335668] pci_bus 0000:05: resource 0 [io 0xb000-0xbfff]
1.335669] pci_bus 0000:05: resource 1 [mem 0xfcc00000-0xfccfffff]
1.335670] pci_bus 0000:05: resource 2 [mem 0xe7300000-0xe73fffff 64bit pref]
1.335671] pci_bus 0000:06: resource 1 [mem 0xfc000000-0xfc0fffff]
1.335671] pci_bus 0000:06: resource 2 [mem 0xe6f00000-0xe6ffffff 64bit pref]
1.335672] pci_bus 0000:07: resource 1 [mem 0xfc400000-0xfc4fffff]
1.335673] pci_bus 0000:07: resource 2 [mem 0xe6b00000-0xe6bfffff 64bit pref]
1.335674] pci_bus 0000:08: resource 1 [mem 0xfc000000-0xfc0fffff]
1.335675] pci_bus 0000:08: resource 2 [mem 0xe6700000-0xe67fffff 64bit pref]
1.335675] pci_bus 0000:09: resource 1 [mem 0xfbc00000-0xfbcfffff]
1.335676] pci_bus 0000:09: resource 2 [mem 0xe6300000-0xe63fffff 64bit pref]
1.335677] pci_bus 0000:0a: resource 1 [mem 0xfb000000-0xfb0fffff]
1.335678] pci_bus 0000:0a: resource 2 [mem 0xe5f00000-0xe5ffffff 64bit pref]
1.335678] pci_bus 0000:0b: resource 0 [io 0x4000-0x4fff]
1.335679] pci_bus 0000:0b: resource 1 [mem 0xfd300000-0xfd3fffff]
1.335680] pci_bus 0000:0b: resource 2 [mem 0xe7a00000-0xe7afffff 64bit pref]
1.335681] pci_bus 0000:0c: resource 0 [io 0x8000-0x8fff]

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1.336141] pci 0000:02:00:0: CLS mismatch (32 != 64), using 64 bytes
1.338887] PCI-DMA: Using software bounce buffering for IO (SWIOTLB)
1.372986] pcieport 0000:00:15:0: PME: Signaling with IRQ 24
1.373095] pcieport 0000:00:15:0: pciehp: Slot #160 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.374313] pcieport 0000:00:15:1: PME: Signaling with IRQ 25
1.374395] pcieport 0000:00:15:1: pciehp: Slot #161 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.375519] pcieport 0000:00:15:2: PME: Signaling with IRQ 26
1.375597] pcieport 0000:00:15:2: pciehp: Slot #162 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.376850] pcieport 0000:00:15:3: PME: Signaling with IRQ 27
1.376916] pcieport 0000:00:15:3: pciehp: Slot #163 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.378089] pcieport 0000:00:15:4: PME: Signaling with IRQ 28
1.378219] pcieport 0000:00:15:4: pciehp: Slot #164 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.379572] pcieport 0000:00:15:5: PME: Signaling with IRQ 29
1.379654] pcieport 0000:00:15:5: pciehp: Slot #165 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.380686] pcieport 0000:00:15:6: PME: Signaling with IRQ 30
1.380781] pcieport 0000:00:15:6: pciehp: Slot #166 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.382046] pcieport 0000:00:15:7: PME: Signaling with IRQ 31
1.382111] pcieport 0000:00:15:7: pciehp: Slot #167 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.383478] pcieport 0000:00:16:0: PME: Signaling with IRQ 32
1.383546] pcieport 0000:00:16:0: pciehp: Slot #192 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.384537] pcieport 0000:00:16:1: PME: Signaling with IRQ 33
1.384600] pcieport 0000:00:16:1: pciehp: Slot #193 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.385725] pcieport 0000:00:16:2: PME: Signaling with IRQ 34
1.385824] pcieport 0000:00:16:2: pciehp: Slot #194 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.387142] pcieport 0000:00:16:3: PME: Signaling with IRQ 35
1.387236] pcieport 0000:00:16:3: pciehp: Slot #195 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
D+
1.409441] pcieport 0000:00:18:5: PME: Signaling with IRQ 53
1.409505] pcieport 0000:00:18:5: pciehp: Slot #261 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.410548] pcieport 0000:00:18:6: PME: Signaling with IRQ 54
1.410611] pcieport 0000:00:18:6: pciehp: Slot #262 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.411706] pcieport 0000:00:18:7: PME: Signaling with IRQ 55
1.411870] pcieport 0000:00:18:7: pciehp: Slot #263 AttnBtn+ PwrCtrl+ MRL- AttnInd- PwrInd- HotPlug+ Surprise- Interlock- NoCompl+ IbPresDis- LLActRe
D+
1.457791] ehci-pci 0000:02:04:0: EHCI Host Controller
1.457796] ehci-pci 0000:02:04:0: new USB bus registered, assigned bus number 2
1.458204] ehci-pci 0000:02:04:0: irq 18, to mem 0xfd5ee000
1.464471] ehci-pci 0000:02:04:0: USB 2.0 started, EHCI 1.00
1.683964] pcieport 0000:00:16:3: hash matches
1.850835] input: VMware VMware Virtual USB Mouse as /devices/pci0000:00/0000:00:15:0/0000:03:00:0/usb3/3-1/3-1:1.0/0003:0E0F:0003.0001/input/input
2.330684] e1000 0000:02:00:0 eth0: (PCI:66MHz:32-bit) 00:0c:29:b3:4e:7b

```

Assess Storage Health and Device-Specific Capabilities

device model : VMware Virtual S & VMware Virtual SATA

CDRW Drive

rotational or non-rotational status : rotational

health summary : unavailable

temperature or warning fields if available : 0C, tapi tidak valid

VMware virtual S berpasangan dengan controller Broadcom / LSI 53c1030 PCI-X Fusion-MPT Dual Ultra320 SCSI. VMware Virtual SATA CDRW dengan controller VMware SATA AHCI

```
fakhrusy@ubuntudesktop01:~$ lsblk -d -o NAME,MODEL,SIZE,ROTA,TYPE
NAME      MODEL                                SIZE ROTA TYPE
loop0                                           4K    1 loop
loop1                                           74M    1 loop
loop2                                           66.8M  1 loop
loop3                                           251.7M  1 loop
loop4                                           66.8M  1 loop
loop5                                           531.4M  1 loop
loop6                                           254.9M  1 loop
loop7                                           18.5M  1 loop
loop8                                           16.4M  1 loop
loop9                                           606.1M  1 loop
loop10                                          395M    1 loop
loop11                                           91.7M  1 loop
loop12                                           10.8M  1 loop
loop13                                           48.1M  1 loop
loop14                                           227.1M  1 loop
loop15                                           576K    1 loop
loop16                                           227.1M  1 loop
loop17                                           48.4M  1 loop
loop18                                           580K    1 loop
loop19                                           15.5M  1 loop
sda        VMware Virtual S                     40G    1 disk
sr0        VMware Virtual SATA CDRW Drive      6.2G    1 rom
```

```
fakhrusy@ubuntudesktop01:~$ sudo smartctl -a /dev/sda
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-6.17.0-19-generic] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:                VMware,
Product:               VMware Virtual S
Revision:              1.0
User Capacity:         42,949,672,960 bytes [42.9 GB]
Logical block size:    512 bytes
Rotation Rate:         Solid State Device
Device type:           disk
Local Time is:         Sun Apr  5 20:59:11 2026 WIB
SMART support is:      Unavailable - device lacks SMART capability.

=== START OF READ SMART DATA SECTION ===
Current Drive Temperature:     0 C
Drive Trip Temperature:        0 C

Error Counter logging not supported

Device does not support Self Test logging
```

Create a Cross-Tool Hardware Validation Report

System Identity

```
fakhrusy@ubuntudesktop01:~$ hostnamectl
Static hostname: ubuntudesktop01
    Icon name: computer-vm
    Chassis: vm 🖥️
    Machine ID: 36452fdc357d4813a1449755d5ec9a03
    Boot ID: 02a612b5f0774e08aaded49d9b899091
Virtualization: vmware
Operating System: Ubuntu 24.04.4 LTS
    Kernel: Linux 6.17.0-19-generic
    Architecture: x86-64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
    Firmware Date: Thu 2020-11-12
    Firmware Age: 5y 4month 3w 1d
```


CPU Topology

```
fakhrusy@ubuntudesktop01:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          45 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 4
On-line CPU(s) list:    0-3
Vendor ID:              GenuineIntel
Model name:             12th Gen Intel(R) Core(TM) i7-12700H
CPU family:             6
Model:                  154
Thread(s) per core:     1
Core(s) per socket:     2
Socket(s):              2
Stepping:               3
BogoMIPS:               5375.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss h
                        t syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_
                        tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave
                        avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi
                        1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves av
                        x_vnni arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_l1d arch_c
                        apabilities
Virtualization features:
Hypervisor vendor:      VMware
Virtualization type:    full
Caches (sum of all):
L1d:                    192 KiB (4 instances)
L1i:                    128 KiB (4 instances)
L2:                     5 MiB (4 instances)
L3:                     48 MiB (2 instances)
NUMA:
```

```
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-3
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:          Not affected
L1tf:                   Mitigation; PTE Inversion
Mds:                    Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown:               Mitigation; PTI
Mmio stale data:        Not affected
Old microcode:          Not affected
Reg file data sampling:  Vulnerable; No microcode
Retbleed:               Mitigation; IBRS
Spec rstack overflow:   Not affected
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:             Mitigation; IBRS; IBPB conditional; STIBP disabled; RSB filling; PBRSB-eIBRS Not affected; BHI SW loop
                        , KVM SW loop
Srbds:                  Not affected
Tsa:                    Not affected
Tsx async abort:        Not affected
Vmscape:                Not affected
```

CPU Flags

```
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss h
                        t syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology tsc_reliable nonstop_
                        tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave
                        avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi
                        1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves av
                        x_vnni arat umip gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_l1d arch_c
                        apabilities
```

Cache Hierarchy

```
fakhrusy@ubuntudesktop01:~$ lscpu | grep -i cache
L1d cache: 192 KiB (4 instances)
L1i cache: 128 KiB (4 instances)
L2 cache: 5 MiB (4 instances)
L3 cache: 48 MiB (2 instances)
```

Physical memory, runtime memory, swap

```
fakhrusy@ubuntudesktop01:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        1.4Gi        4.2Gi         45Mi        2.4Gi        6.4Gi
Swap:         4.0Gi          0B        4.0Gi
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/meminfo
```

```
MemTotal:      8083272 kB
MemFree:       4438972 kB
MemAvailable:  6666728 kB
Buffers:       57292 kB
Cached:        2397328 kB
SwapCached:    0 kB
Active:        2245204 kB
Inactive:      785584 kB
Active(anon):  622588 kB
Inactive(anon): 0 kB
Active(file):  1622616 kB
Inactive(file): 785584 kB
Unevictable:   16 kB
Mlocked:       16 kB
SwapTotal:     4194300 kB
SwapFree:      4194300 kB
Zswap:         0 kB
Zswapped:      0 kB
```

```
fakhrusy@ubuntudesktop01:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        1.4Gi        4.2Gi         45Mi        2.4Gi        6.4Gi
Swap:         4.0Gi          0B        4.0Gi
```

```
fakhrusy@ubuntudesktop01:~$ vmstat -s
```

```
8083272 K total memory
1420528 K used memory
2247856 K active memory
785632 K inactive memory
4434940 K free memory
57340 K buffer memory
2477800 K swap cache
4194300 K total swap
0 K used swap
4194300 K free swap
3973 non-nice user cpu ticks
226 nice user cpu ticks
3445 system cpu ticks
806540 idle cpu ticks
492 IO-wait cpu ticks
0 IRQ cpu ticks
166 softirq cpu ticks
0 stolen cpu ticks
0 non-nice guest cpu ticks
0 nice guest cpu ticks
2012587 K paged in
227276 K paged out
```

Storage map

```
fakhrusy@ubuntudesktop01:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0     4K  1 loop /snap/bare/5
loop1       7:1      0    74M  1 loop /snap/core22/2292
loop2       7:2      0   66.8M  1 loop /snap/core24/1499
loop3       7:3      0  251.7M  1 loop /snap/firefox/7766
loop4       7:4      0   66.8M  1 loop /snap/core24/1587
loop5       7:5      0  531.4M  1 loop /snap/gnome-42-2204/247
loop6       7:6      0  254.9M  1 loop /snap/firefox/7967
loop7       7:7      0   18.5M  1 loop /snap/firmware-updater/212
loop8       7:8      0   16.4M  1 loop /snap/firmware-updater/223
loop9       7:9      0  606.1M  1 loop /snap/gnome-46-2404/153
loop10      7:10     0   395M  1 loop /snap/mesa-2404/1165
loop11      7:11     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop12      7:12     0   10.8M  1 loop /snap/snap-store/1270
loop13      7:13     0   48.1M  1 loop /snap/snapd/25935
loop14      7:14     0  227.1M  1 loop /snap/thunderbird/1040
loop15      7:15     0    576K  1 loop /snap/snapd-desktop-integration/343
loop16      7:16     0  227.1M  1 loop /snap/thunderbird/1029
loop17      7:17     0   48.4M  1 loop /snap/snapd/26382
loop18      7:18     0    580K  1 loop /snap/snapd-desktop-integration/357
loop19      7:19     0   15.5M  1 loop /snap/snap-store/1310
sda         8:0      0   40G   0 disk
├─sda1      8:1      0     1M   0 part
└─sda2      8:2      0   40G   0 part /
sr0        11:0     1   6.2G   0 rom  /media/fakhrusy/Ubuntu 24.04.4 LTS amd641
```

PCI devices

```
fakhrusy@ubuntudesktop01:~$ lspci
00:00.0 Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge (rev 01)
00:01.0 PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge (rev 01)
00:07.0 ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA (rev 08)
00:07.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:07.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:07.7 System peripheral: VMware Virtual Machine Communication Interface (rev 10)
00:0f.0 VGA compatible controller: VMware SVGA II Adapter
00:10.0 SCSI storage controller: Broadcom / LSI 53c1030 PCI-X Fusion-MPT Dual Ultra320 SCSI (rev 01)
00:11.0 PCI bridge: VMware PCI bridge (rev 02)
00:15.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.1 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.2 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.3 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.4 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.5 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.6 PCI bridge: VMware PCI Express Root Port (rev 01)
00:15.7 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.1 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.2 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.3 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.4 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.5 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.6 PCI bridge: VMware PCI Express Root Port (rev 01)
00:16.7 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.0 PCI bridge: VMware PCI Express Root Port (rev 01)
```



```

00:17.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.1 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.2 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.3 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.4 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.5 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.6 PCI bridge: VMware PCI Express Root Port (rev 01)
00:17.7 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.0 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.1 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.2 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.3 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.4 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.5 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.6 PCI bridge: VMware PCI Express Root Port (rev 01)
00:18.7 PCI bridge: VMware PCI Express Root Port (rev 01)
02:00.0 Ethernet controller: Intel Corporation 82545EM Gigabit Ethernet Controller (Copper) (rev 01)
02:01.0 SATA controller: VMware SATA AHCI controller
02:02.0 USB controller: VMware USB1.1 UHCI Controller
02:03.0 Multimedia audio controller: Ensoniq ES1371/ES1373 / Creative Labs CT2518 (rev 02)
02:04.0 USB controller: VMware USB2 EHCI Controller
03:00.0 USB controller: VMware Device 077a

```

USB devices

```

fakhrusy@ubuntu01:~$ lsusb
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub

```

Network hardware

```

fakhrusy@ubuntu01:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:b3:4e:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s0
fakhrusy@ubuntu01:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b3:4e:7b brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.150.128/24 brd 192.168.150.255 scope global dynamic noprefixroute ens32
        valid_lft 1183sec preferred_lft 1183sec
    inet6 fe80::20c:29ff:feb3:4e7b/64 scope link
        valid_lft forever preferred_lft forever

```

Kernel-detected events

```
fakhrusy@ubuntu01:~$ sudo dmesg | grep -iE 'warn|error|fail|bug'
[ 0.686653] ACPI: [Firmware Bug]: BIOS _OSI(Linux) query ignored
[ 0.700259] PCI: Using host bridge windows from ACPI; if necessary, use "pci=nocrs" and report a bug
[ 1.332592] pci 0000:00:15.0: bridge window [io size 0x1000]: failed to assign
[ 1.332593] pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
[ 1.332595] pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
[ 1.332597] pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
[ 1.332598] pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
[ 1.332600] pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
[ 1.332601] pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
[ 1.332603] pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
[ 1.332604] pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
[ 1.332606] pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
[ 1.332608] pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
[ 1.332609] pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
[ 1.332611] pci 0000:00:17.4: bridge window [io size 0x1000]: failed to assign
[ 1.332612] pci 0000:00:17.5: bridge window [io size 0x1000]: failed to assign
[ 1.332614] pci 0000:00:17.6: bridge window [io size 0x1000]: failed to assign
[ 1.332615] pci 0000:00:17.7: bridge window [io size 0x1000]: failed to assign
[ 1.332617] pci 0000:00:18.3: bridge window [io size 0x1000]: failed to assign
[ 1.332619] pci 0000:00:18.4: bridge window [io size 0x1000]: failed to assign
[ 1.332620] pci 0000:00:18.5: bridge window [io size 0x1000]: failed to assign
[ 1.332622] pci 0000:00:18.6: bridge window [io size 0x1000]: failed to assign
[ 1.332623] pci 0000:00:18.7: bridge window [io size 0x1000]: failed to assign
[ 1.332627] pci 0000:00:18.7: bridge window [io size 0x1000]: failed to assign
[ 1.332636] pci 0000:00:17.6: bridge window [io size 0x1000]: failed to assign
[ 1.332638] pci 0000:00:17.5: bridge window [io size 0x1000]: failed to assign
[ 1.332640] pci 0000:00:17.4: bridge window [io size 0x1000]: failed to assign
[ 1.332641] pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
[ 1.332643] pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
[ 1.332644] pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
[ 1.332646] pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
[ 1.332648] pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
[ 1.332649] pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
[ 1.332651] pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
[ 1.332652] pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
[ 1.332654] pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
[ 1.332655] pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
[ 1.332657] pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
[ 1.332659] pci 0000:00:15.0: bridge window [io size 0x1000]: failed to assign
[ 1.686264] RAS: Correctable Errors collector initialized.
[ 4.286262] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File System...
[ 4.322723] systemd[1]: Mounted sys-kernel-debug.mount - Kernel Debug File System.
[ 9.884493] WARNING: CPU: 0 PID: 773 at drivers/gpu/drm/vmwgfx/vmwgfx_bo.c:77 vmw_bo_free+0x41/0x60 [vmwgfx]
```